Cognitio Integralis Recognitionis Navigationisque Ordinatrix

Project CIRNO

A multiagentic study & scheduling platform

Y. Zhan: Development of intelligent agents, RAG databases, and file storage systems
L. Xiong: Backend / conventional database development, construction of cognitive psychology models
J. Lin: Frontend development
Z. Zhou: Backend development

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CODE OF CONDUCT NOTE: THIS PROJECT UTILISES Google Gemini 3.1 Pro FOR FRONTEND DESIGN ASSISTANCE.

BACKGROUND & INSPIRATION 02

CORE KEYPOINT

Unbalancement of students' academic achievement & Physical/mental health

chronic burnout for grades

academic failure of disengagement

INSPIRATION

Balance life & study

Dynamic & Science framework

CORE POSITIONING 03

An LLM + cognitive psychology tool;
A personalized learning platfrom;

Lack of personalized study planning & unbalanced cognitive load

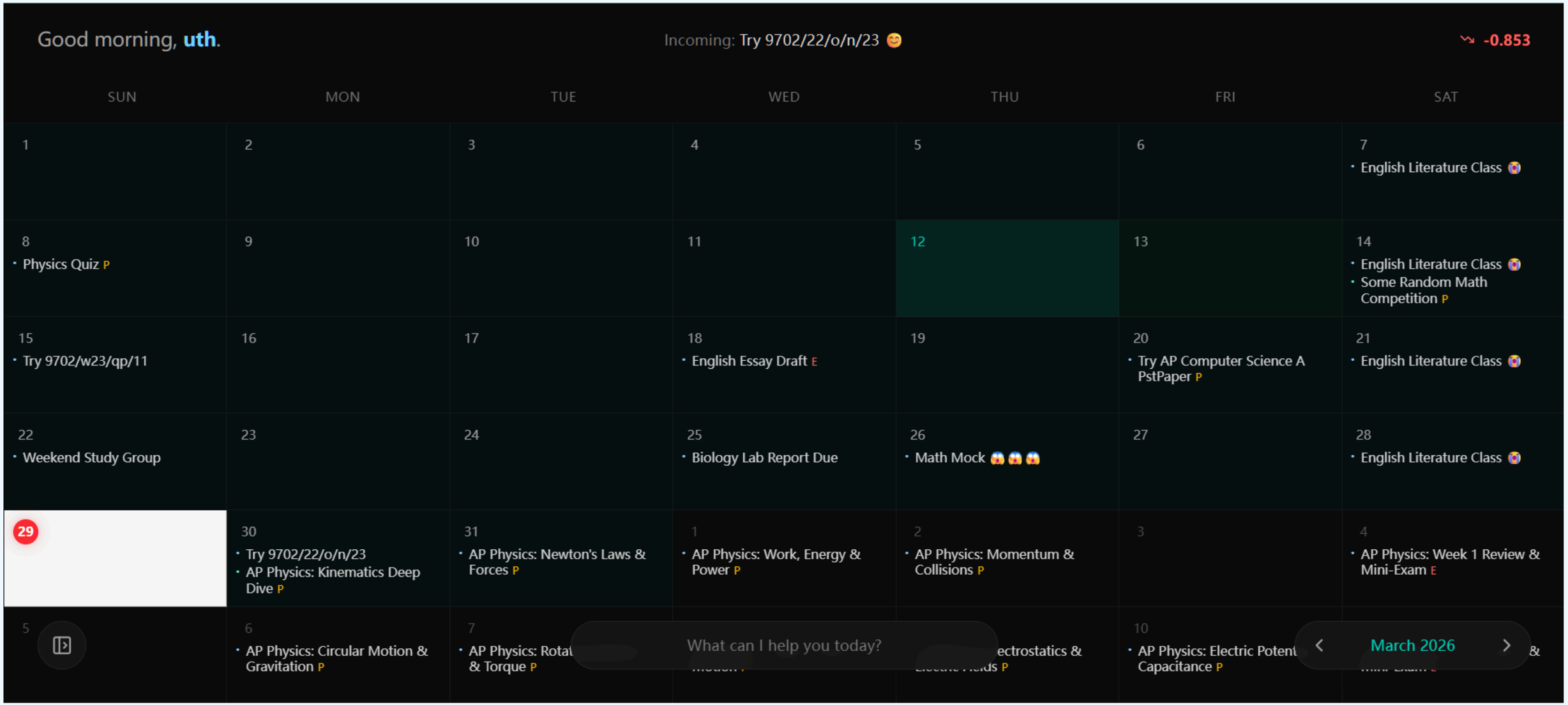
Difficulty in identifying knowledge gaps and unfocused revision

Lack of authoritative information in Q&A and difficulty accessing cutting-edge research insights

Scattered error recording that cannot feed back into study planning

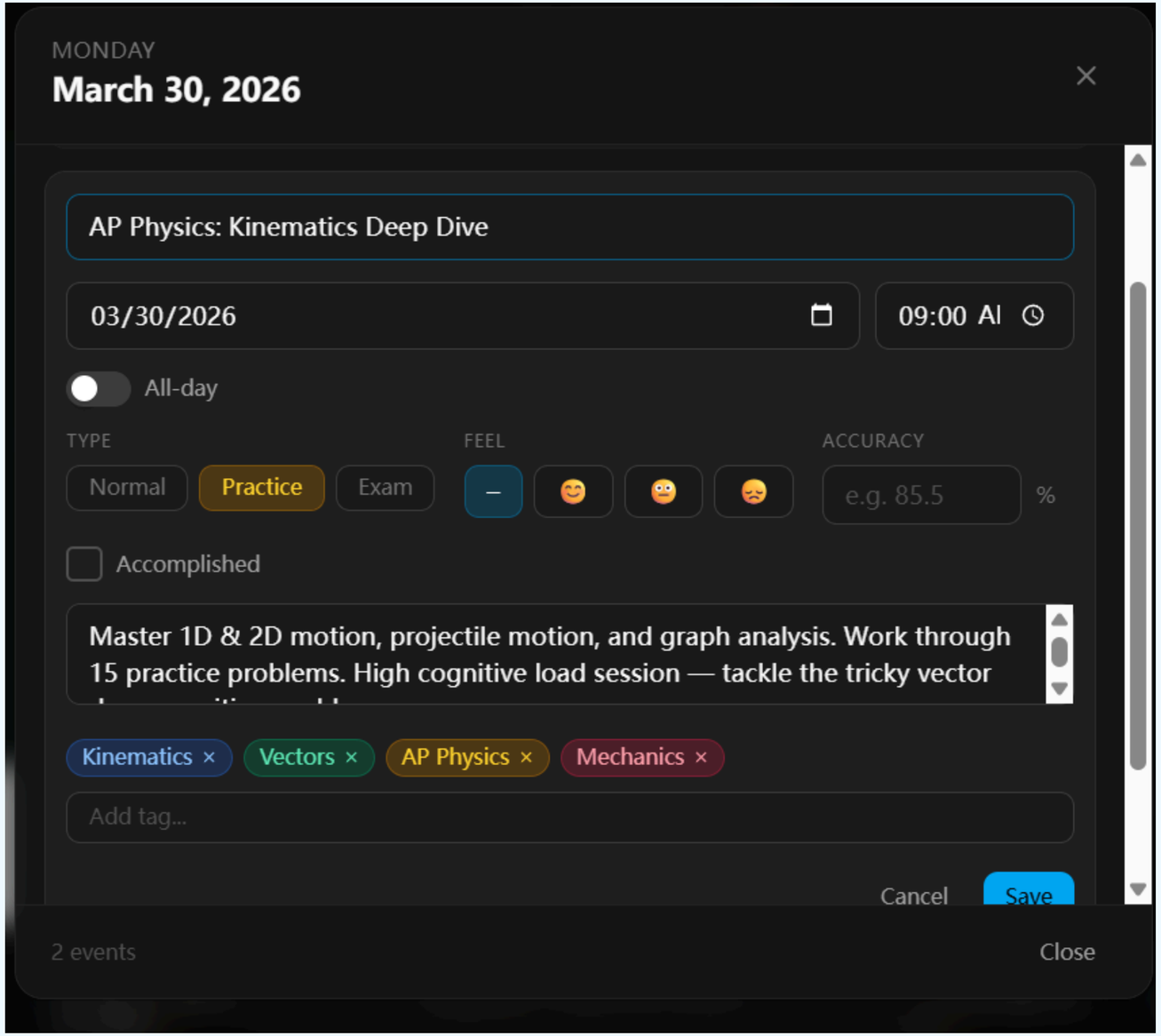
Created, trying to mitigate dichotomy of learning unhappily and happying unlearningly

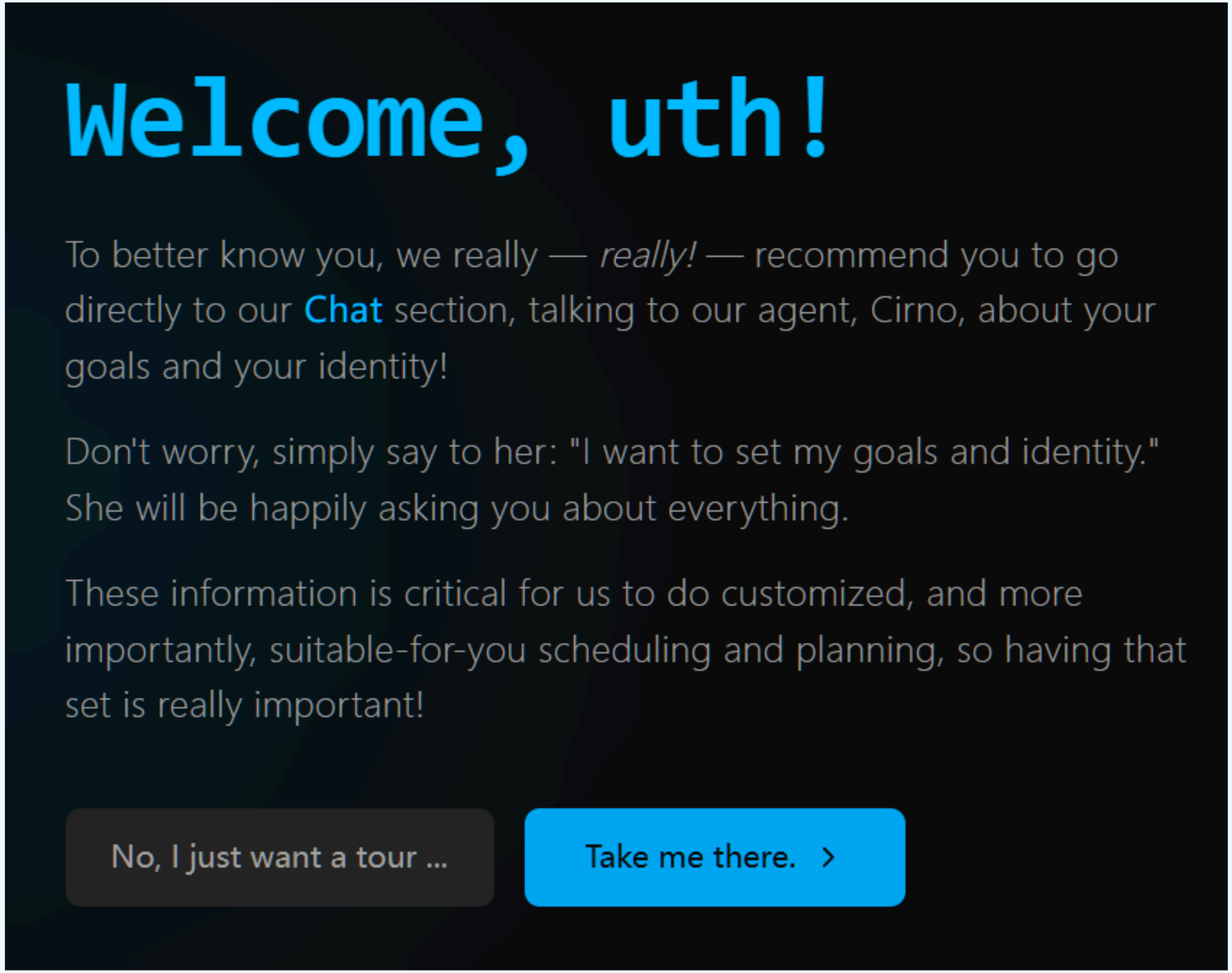
Scheduling system



We know you well.
We keep record of you

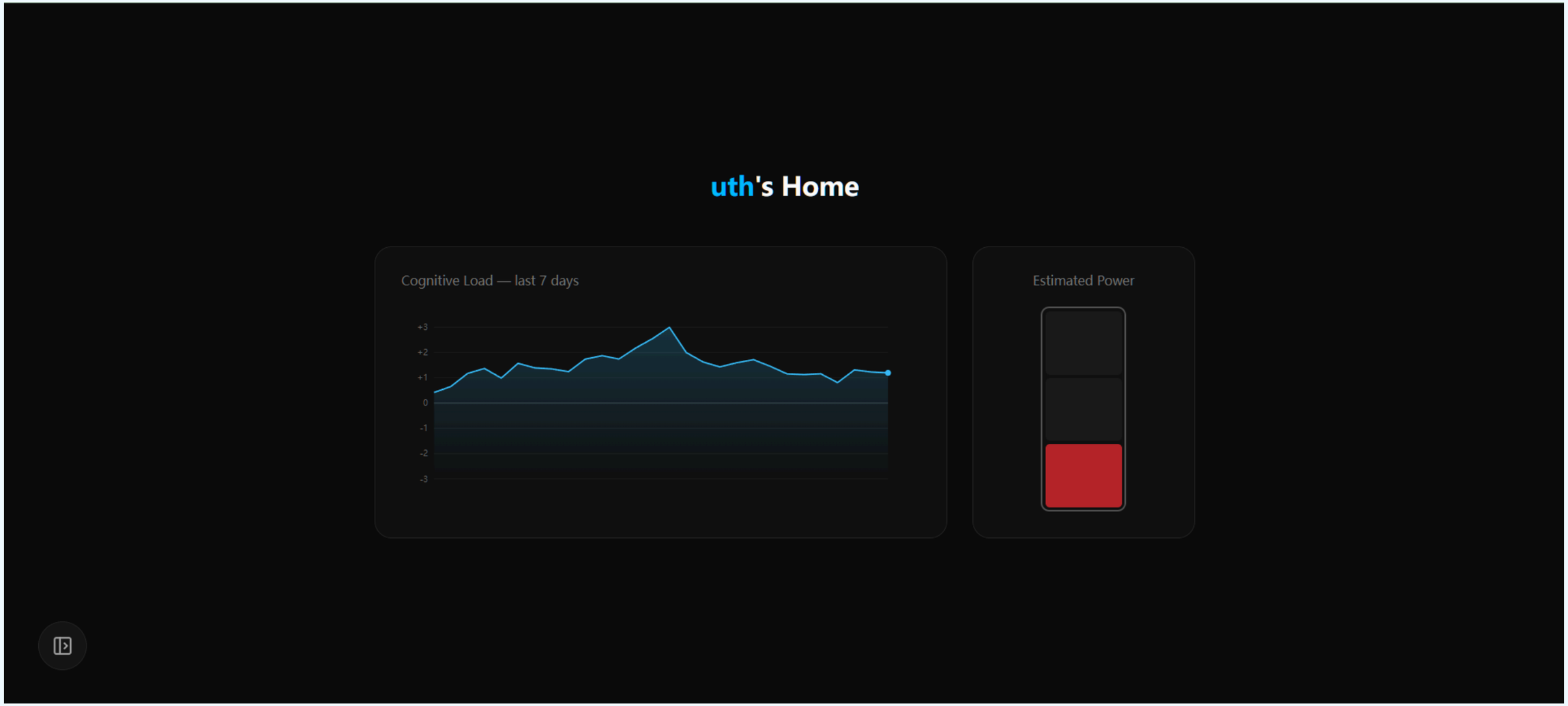
... even if you don’t remember them.



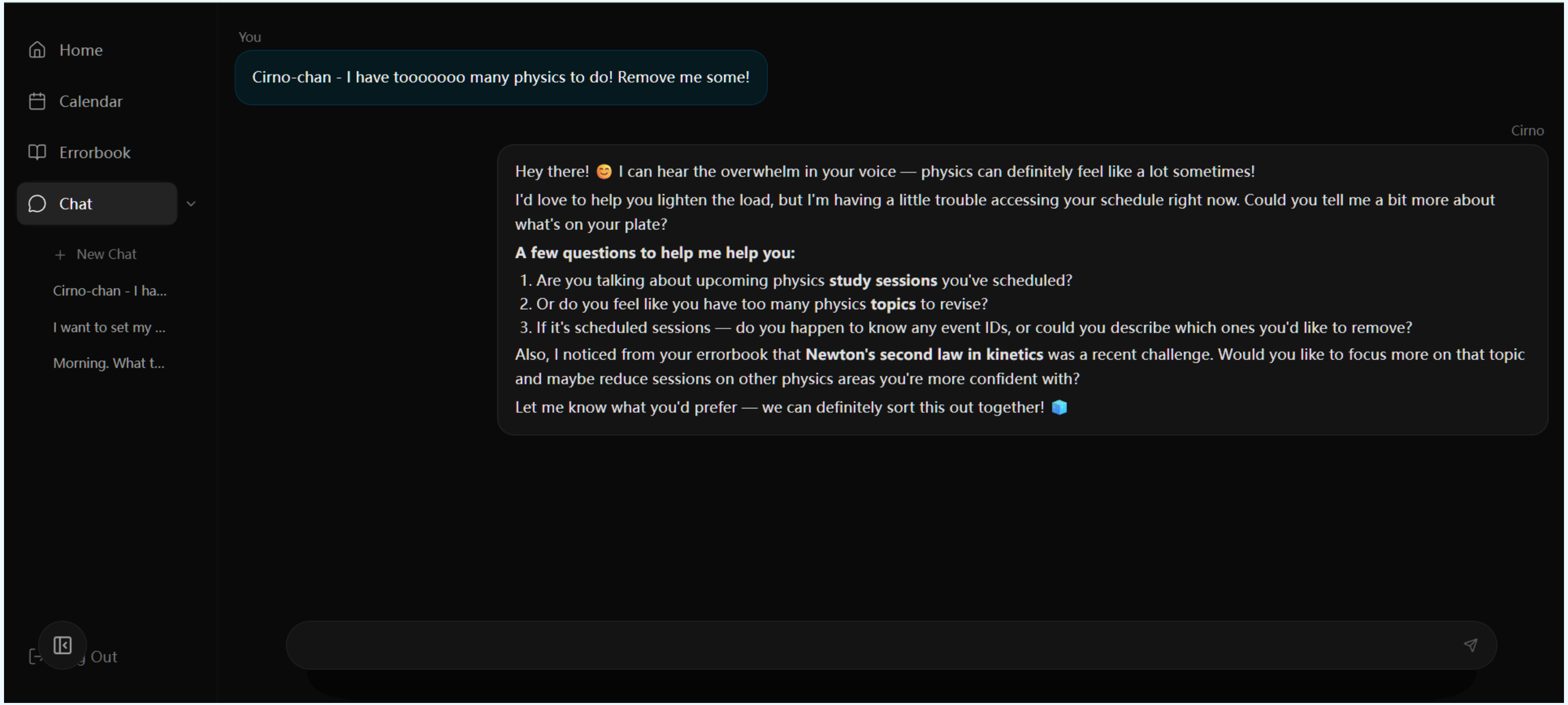


Every individuals matters.

History tracking of status

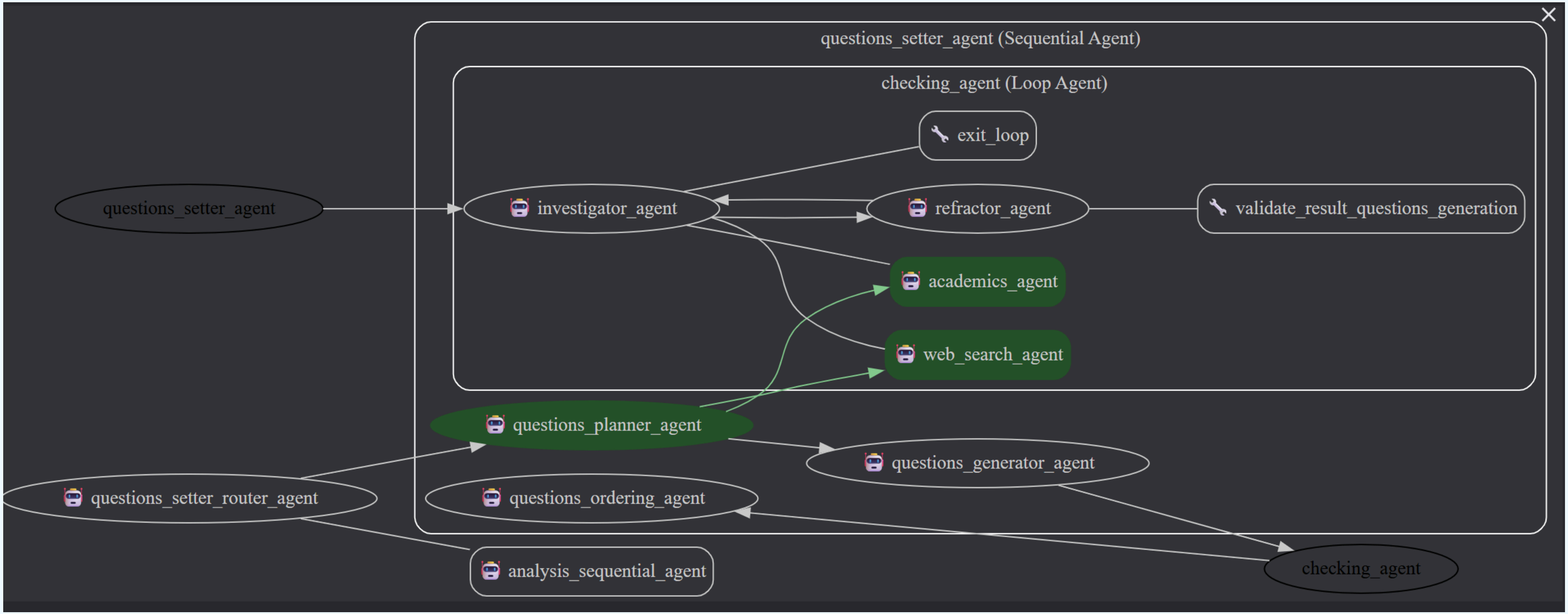


Buddy of study.



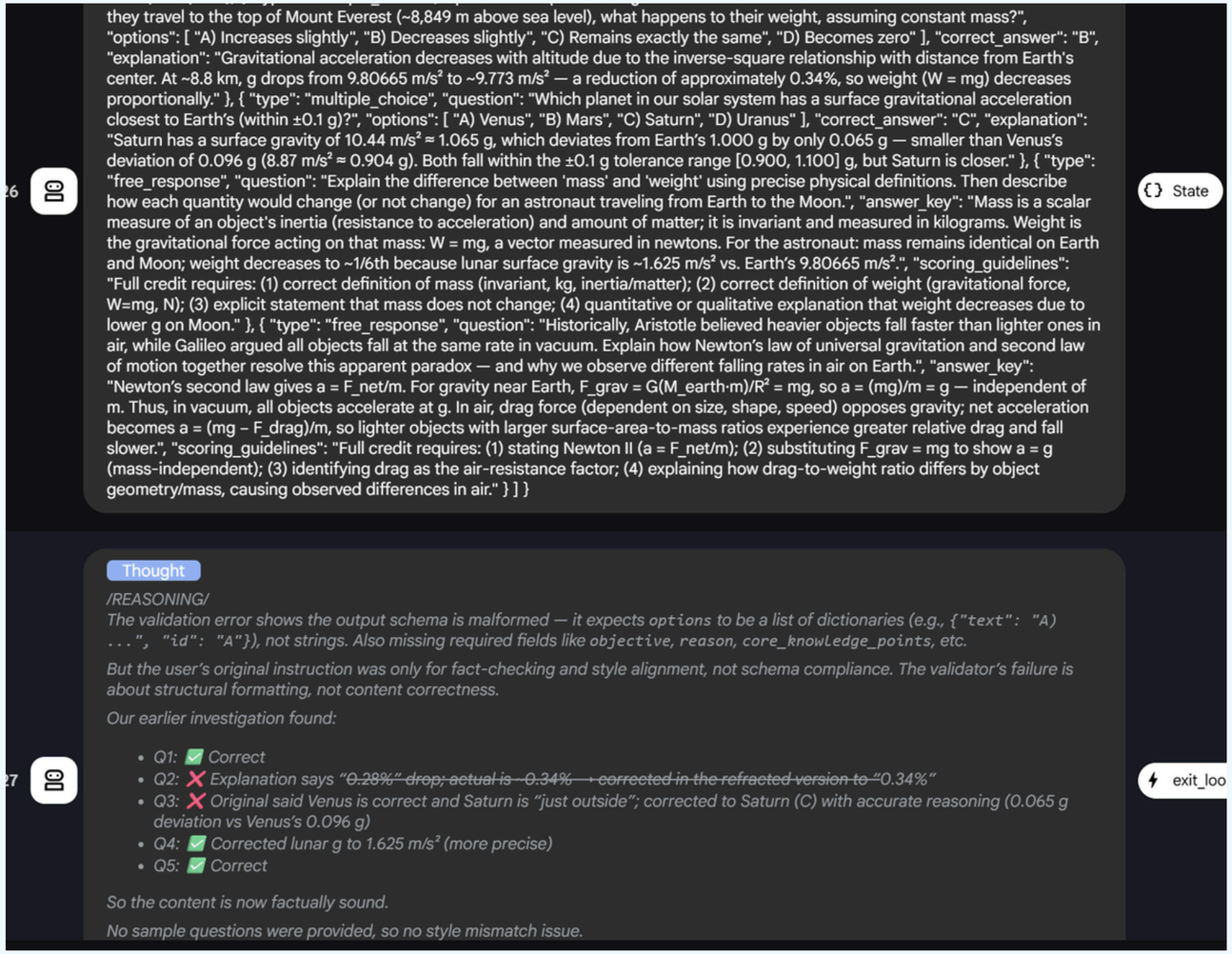
Multi-agent architecture.

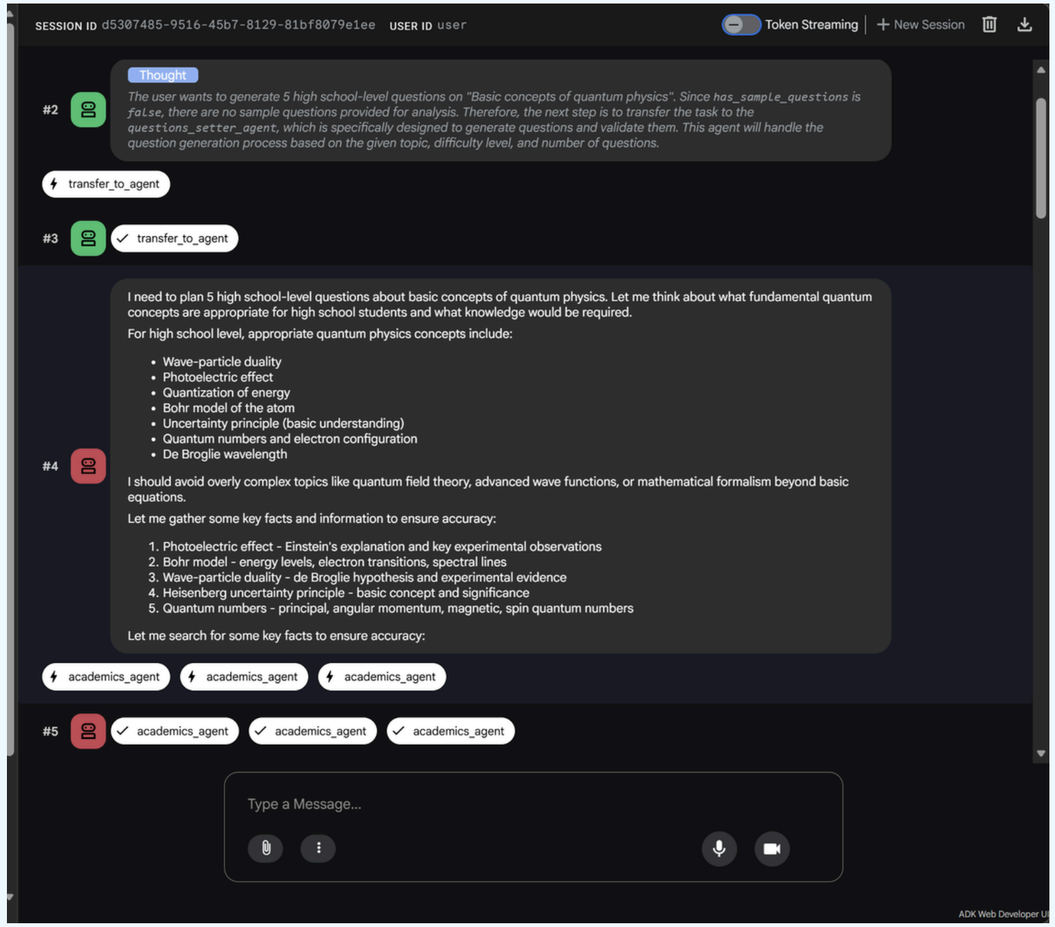
More specialised, more accurate.



Multi-agent architecture.

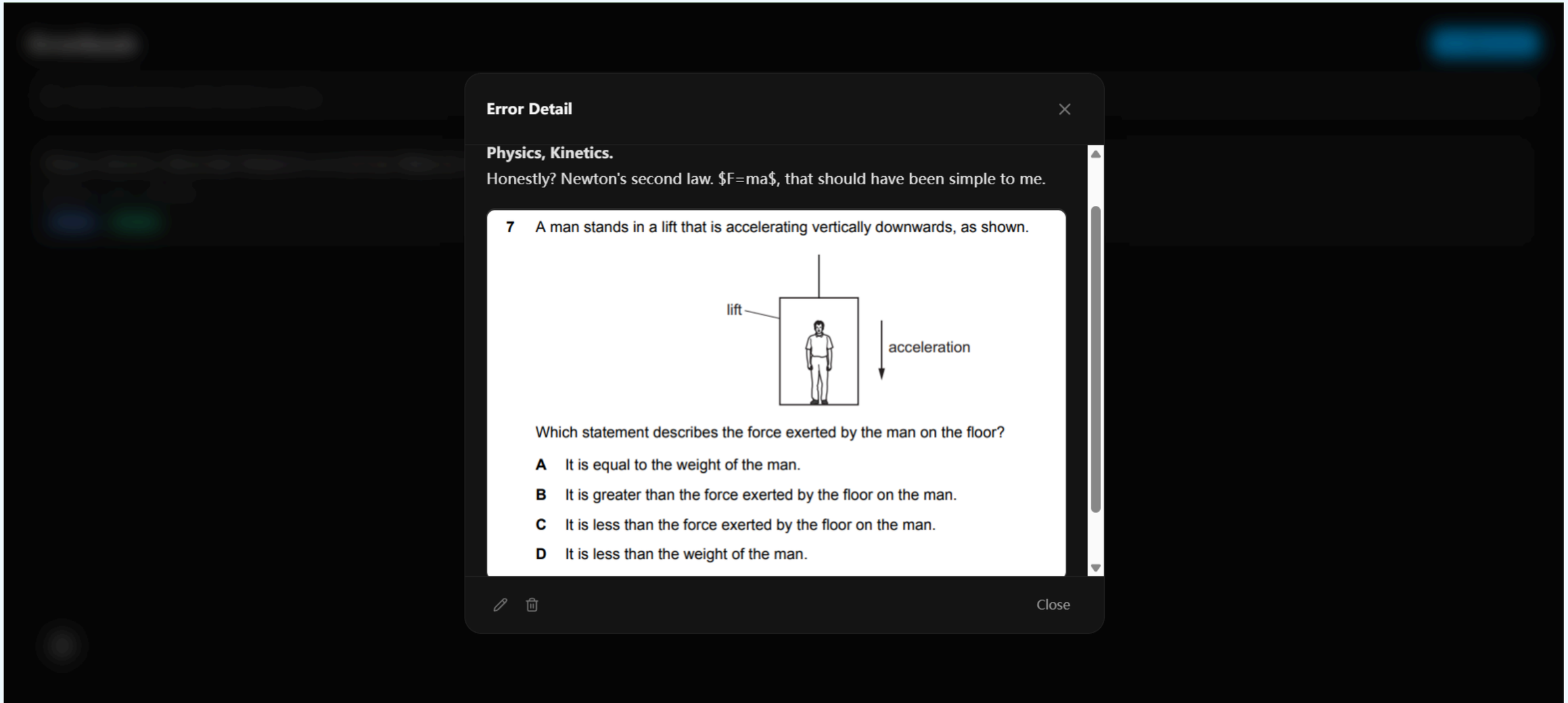
The cooperation between different agents:





Error tracking

Knowing oneself is weak makes one strong.



Paas & Van Merriënboer

Instructional Efficiency Model

$$P_{raw} = w_1 \cdot TestScore + w_2 \cdot (1 - DelayRate)$$

$$L_{raw} = w_3 \cdot CheckIn + w_4 \cdot NLPSeverity$$

$$Z_P = \frac{P_{raw} - \mu_P}{\sigma_P}$$

$$Z_L = \frac{L_{raw} - \mu_L}{\sigma_L}$$

$$Efficiency = \frac{Z_P - Z_L}{\sqrt{2}}$$

Theoretical Basis

Bloom's Taxonomy

A hierarchical framework used to classify educational learning objectives into levels of intellectual complexity.

Future

- Personal file storage and RAG (textbook or other materials).
- Comprehend question management (Use NoSQL Database to store the structured question data with the embedded vector): Question recommendation.
- Vector database to store embedded file chunks and question stems for RAG and analysis.
- FRQ correcting agent.

Frontend: StudentAssistantTeam/project-cirno-backend

Backend: StudentAssistantTeam/project-cirno-backend

Web Search Agent: StudentAssistantTeam/CIRNO_QuestionsGeneratorAgent

Questions Generation agent:StudentAssistantTeam/CIRNO_QuestionsGeneratorAgent

Math and ScienceAgent:StudentAssistantTeam/CIRNO_MathAndScienceAgent

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Feel absolutely free to clone, build and try (before our stable release)
or PR/issue us to improve!